Grade 7 Name_Answer Key________
Unit 11 Date____________________
Lesson 5 Practice Period___________________

Mike earned $\$60{,}000$ per year for the last five years. He contributes a percentage of each paycheck to his retirement fund. Starting at 2%, he set his contribution amount to increase by 2.6% each year. Using this information, complete the table of values.

Year	**Contribution Rate**	**Amount contributed per year**
1	2%	$60{,}000 \cdot 02 = 1{,}200$
2	4.6%	$60{,}000 \cdot .046 = 2{,}760$
3	7.2%	$60{,}000 \cdot .072 = 4{,}320$
4	9.8%	$60{,}000 \cdot .098 = 5{,}880$
5	12.4%	$60{,}000 \cdot .124 = 7{,}440$

1. What year will Mike contribute at least $\$2{,}500$?

   Year 2

2. How much more money is Mike contributing between year 5 and year 2?

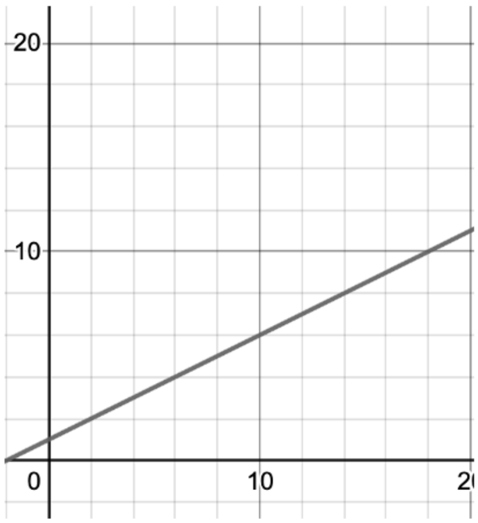
   Year 5: $7{,}440$
   Year 2: $2{,}760$ $7{,}440 - 2{,}760 = \$4{,}680$

3. Explain whether the relationship between contribution percentage and year is or is not proportional.

   It is not proportional as the rate of amount contributed per year is not the same.

Determine if each graph is proportional or not proportional. Circle the correct classification.

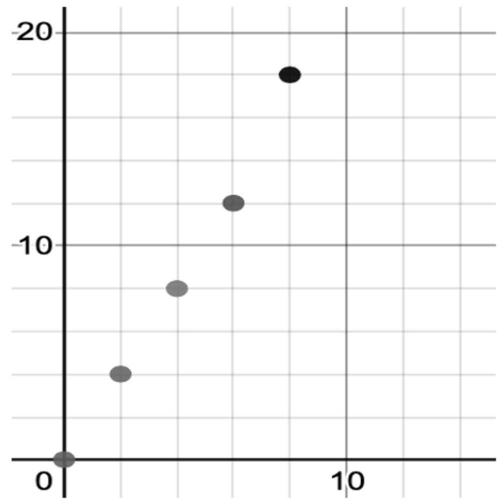
4.



- proportional
- not proportional

not at (0,0)

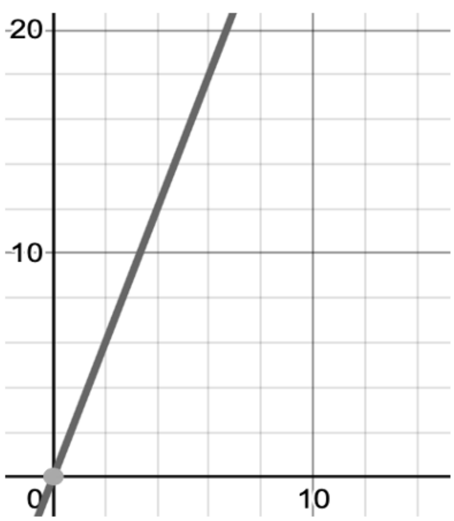
5.



- proportional
- not proportional

(8,18) is not a constant rate of change

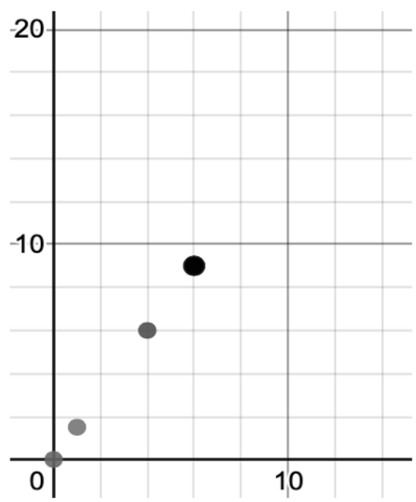
6.



- proportional
- not proportional

(0,0) and constant rate

7.



- proportional
- not proportional

(0,0) and constant rate